

PRODUCT NAME: PRIO

**PRIO: A MULTI-TENANT QR-BASED PRIORITY QUEUE
PLATFORM FOR WALK-IN CUSTOMERS, REMOTE JOINERS,
AND SERVICE TEAMS**

A Capstone Project Presented to the Faculty of

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ABSTRACT

Long physical lines remain a common problem in service-oriented environments such as clinics, school offices, municipal counters, retail shops, and small business service desks. Traditional first-come, first-served arrangements often force customers to wait in crowded areas without clear visibility into queue progress. This situation increases uncertainty, causes missed turns, and places pressure on staff who must manually track the order of service. The project entitled *Prio* addresses this problem through a multi-tenant web platform that digitizes ticket issuance, live queue monitoring, and near-turn customer notifications.

Prio is designed as a tenant-aware queue platform where each vendor or service unit has its own workspace, queue settings, public join link, and public monitor page. Customers may join through a QR code or through an online workflow, while vendors can issue walk-in tickets directly from a dashboard. The system generates ticket numbers per tenant and per day, publishes queue updates in real time, and notifies customers through email or SMS when their turn is approaching or when they are actively called for service. The proposed implementation uses a React and Vite frontend, an Express backend, MongoDB for data persistence, and Server-Sent Events for live queue updates.

The project is positioned as a practical, scalable, and accessible solution for organizations that need modern queue handling without building separate systems for each branch or vendor. Its primary contribution lies in combining multi-tenancy, QR-based self-service queue entry, live public visibility, and configurable near-turn alerts in a single web application. By reducing uncertainty and minimizing overcrowded waiting areas, Prio aims to improve customer experience, support service staff, and provide a useful foundation for future enhancements such as multiple counters, business-hour controls, role separation, and production-grade anti-abuse protections.

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CHAPTER I

INTRODUCTION

Rationale of the Study

Service queues are still managed manually in many small and medium-sized organizations. Customers often line up physically, ask staff for updates, or remain near the service area because they do not know when they will be called. This results in long perceived waiting times, congestion, and inconsistent customer experience. For vendors and staff, manual queue handling also creates unnecessary operational work because ticket numbers, queue status, and customer follow-up must be coordinated through verbal announcements or paper-based tracking.

The growing availability of smartphones and internet-connected devices has made digital queueing more realistic even for smaller organizations. Customers already expect to scan a QR code, receive updates remotely, and monitor service progress through a browser. However, many available queueing solutions are designed as closed commercial platforms or are optimized for a single organization rather than for a platform that can support several independent vendors with isolated settings and queue records. This creates an opportunity for a web-based solution that combines accessibility, clear queue visibility, and tenant-aware management.

The Prio project is proposed to address this gap. It provides each vendor with its own queue prefix, service settings, public join link, QR target, and live monitor page while keeping the platform centralized. Customers may join before arriving or while physically present, and staff can still issue walk-in tickets when necessary. Real-time monitoring and near-turn alerts reduce uncertainty for customers, while the dashboard helps staff control calling, serving, skipping, and queue configuration from a single interface.

The study is significant because it demonstrates how a modern queueing workflow can be implemented using common web technologies and open standards. It also supports the practical needs of organizations that want a more organized service process without maintaining separate systems for each queueing context. By combining QR-based entry, live queue visibility, and notification support, Prio proposes a solution that is both technically feasible and operationally useful.

Review of Related Works

Existing queue management platforms show that digital queueing can substantially improve transparency and customer experience. Qminder highlights a central service dashboard where staff can manage queues, view visitor details, and notify customers when their turn is ready [1]. QLESS similarly promotes removing the need for customers to wait in a physical line by combining line

management, appointments, and queue intelligence [2]. Queue-Fair, while focused on virtual waiting rooms for high-demand scenarios, emphasizes fairness, estimated wait visibility, and controlled queue progress [3]. These systems validate the value of digital queueing and informed waiting.

Academic work also supports the need for improved queue management. Ndiaye et al. discussed how electronic queue systems can reduce the negative impact of prolonged waiting in public service contexts and argued for more intelligent queue handling in environments where vulnerable users and high service demand are present [4]. This aligns with the practical motivation of Prio, which aims to reduce crowding, uncertainty, and missed turns through a web-based priority number system.

Despite the strengths of existing systems, a gap remains for a platform that is both tenant-aware and lightweight enough for web-first deployment in local service environments. Many commercial platforms focus on enterprise use cases, formal appointment systems, or high-volume access control. Prio instead concentrates on a simpler but highly practical combination of features: vendor-specific queue workspaces, QR-based self-entry, customer lookup codes, public live queue monitoring, and configurable near-turn alerts.

Another relevant technical consideration is live queue synchronization. The WHATWG HTML specification defines Server-Sent Events as a browser-friendly mechanism for streaming updates from server to client over standard HTTP connections [5]. This model is well suited to the project because it enables a public queue monitor and vendor dashboard to reflect queue movement in real time without requiring more complex websocket infrastructure. The selected approach supports the project goal of accessibility and straightforward deployment.

Project Objectives

The general objective of the study is to design and develop a multi-tenant web-based priority queue management system that enables vendors to manage walk-in and remote customers through QR-based queue entry, real-time queue monitoring, and near-turn customer notifications.

The specific objectives of the project are the following:

1. To provide vendor registration and tenant workspace creation so that each service unit can maintain its own queue identity, settings, and access controls.
2. To allow customers to join a queue either by scanning a QR code or by using an online join page, with optional contact details for notifications.
3. To generate queue tickets with tenant-specific prefixes, daily sequence numbers, and lookup codes for easier ticket tracking.

4. To develop a vendor dashboard that supports issuing walk-in tickets, calling the next customer, serving or skipping the current ticket, and updating queue settings.
5. To provide a live public queue board and a ticket-specific monitoring view that update in real time as the queue changes.
6. To send near-turn and called-turn notifications through email or SMS when supported provider credentials are available.
7. To store tenant, user, ticket, and counter data in a persistent database that supports queue history and daily queue sequencing.

Project Scope and Limitation

The scope of Prio covers a browser-based queue management workflow for multiple vendors or service units within one platform. The system includes vendor onboarding, customer registration, authenticated tenant dashboards, QR-based queue entry, online queue joining, ticket generation, queue history, public queue monitoring, and notification preferences for email or SMS alerts. It also includes tenant settings such as queue prefix, average service time, and notification threshold.

The project is limited to one active called ticket per tenant at a given time and follows a first-in, first-out queue order within each tenant. The current implementation is optimized for web access and does not provide a separate native mobile application. Although the notification workflow supports email and SMS channels, dependable production delivery still depends on valid external SMTP and messaging credentials as well as network availability.

In its current form, the project does not yet include advanced production features such as appointment scheduling, multiple simultaneous counters per tenant, role separation between owner and staff accounts, payment integration, anti-spam protection such as CAPTCHA, or full automated test coverage. These limitations are acceptable for the present capstone phase because the main focus of the project is to validate the core queue lifecycle and tenant-aware workflow rather than to deliver a complete enterprise deployment package.

Architectural Diagram

Figure 1 presents the high-level architecture of the proposed system. The design separates the customer and vendor interfaces from the backend queue services while preserving live data flow to public and authenticated pages. MongoDB stores tenant, user, ticket, and counter data, while the backend also coordinates notifications and real-time event publication.

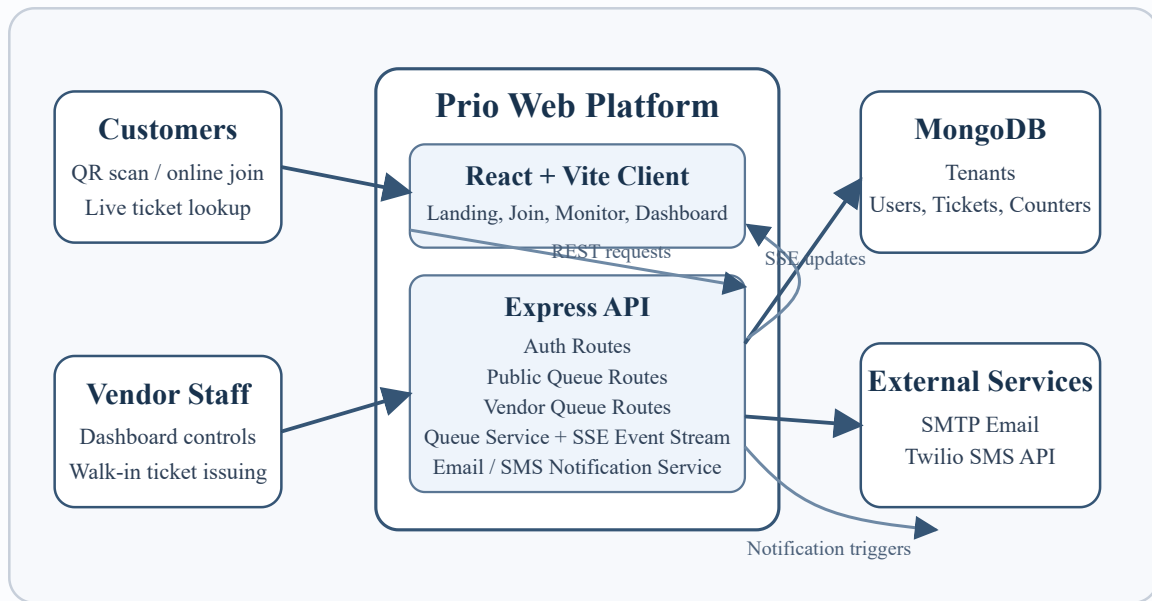


Figure 1. High-level architecture of the Prio queue platform.

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